

Session 13 (AIML) - Assignment Question

Q1. (Basic Array Creation)

Write a program to:

- a) Create a 1D array with elements [10, 20, 30, 40, 50] using `np.array()`.
- b) Create a 2D array (3×3) using `np.array()`.

Print both arrays along with their shape and data type (`dtype`).

Q2. (`np.zeros()` and `np.ones()`)

Create the following using NumPy:

- a) A 1D array of 8 zeros.
- b) A 2D array of shape (4, 4) filled with ones.
- c) A 3×3 matrix of zeros.

Print all arrays with proper labels.

Q3. (`np.arange()`)

Use `np.arange()` to create and print the following arrays:

- a) Numbers from 0 to 20 (step 1).
 - b) Even numbers from 10 to 50.
 - c) Numbers from 5 to 100 with step of 5.
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Q4. (`np.linspace()`)

Use `np.linspace()` to create:

- a) 10 equally spaced numbers between 0 and 5.
- b) 15 equally spaced numbers between -10 and 10.

Print the arrays and their length.

Q5. (Random Arrays)

Write a program to generate:

- a) A 1D array of 10 random numbers between 0 and 1 using `np.random.rand()`.
- b) A 3×3 matrix of random numbers from standard normal distribution using `np.random.randn()`.

c) A 2D array (4×5) of random integers between 10 and 50 using `np.random.randint()`.

Print all with proper headings.

Q6. (Vectors and Basic Operations)

Create two vectors:

```
v1 = np.array([2, 4, 6, 8])
```

```
v2 = np.array([1, 3, 5, 7])
```

Perform and print:

- Addition
 - Subtraction
 - Element-wise multiplication
 - Dot product
-

Q7. (Matrices and Operations)

Create two 3×3 matrices:

```
A = np.array([[1,2,3],[4,5,6],[7,8,9]])
```

```
B = np.array([[9,8,7],[6,5,4],[3,2,1]])
```

Perform the following:

- Matrix addition
- Matrix multiplication (@ or `np.dot()`)
- Element-wise multiplication (*)

Print the results clearly.

Q8. (Properties of Arrays)

Create a 4×4 matrix of random integers between 1 and 100.

Print:

- Shape
- Dimension (ndim)
- Total number of elements (size)
- Data type

- Minimum and maximum values
-

Q9. (Combined - Random + Reshape + Statistics)

Generate a 1D array of 20 random integers between 1 and 50 using `np.random.randint()`.

- Reshape it into a 4×5 matrix.
 - Print the sum, mean, and standard deviation of the matrix.
 - Find the maximum value in each row.
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Q10. (Mini Project - NumPy Application)

Create a **Simple Statistics Calculator** using NumPy:

Write a program that:

- Asks the user how many numbers they want to generate.
- Generates that many random numbers between 10 and 100 using `np.random.randint()`.
- Stores them in a 1D array.
- Prints:
 - The array
 - Mean, Median, Standard Deviation
 - Minimum and Maximum
 - Reshapes into a 2D array (if possible) and prints row-wise sum.

Use proper comments and exception handling where needed.